- أ٤
- ب٤
- ج٤
- د٤
- هـ٤ (selected)

6

رقم الكمبيوتر *

13

7

الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس) *

SQUT

Will be reviewed

8

Consider the shown graph. Using Kruskal’s algorithm, which is the order of the edges added to the spanning tree? *

- CB,BD,CF,FE,DF
- CB,BD,CF,FE,AC
- CB,BD,CF,FE,AD (selected)

- ( ) CB,BD,CF,FE,AB

Will be reviewed

9

In divide & conquer strategy, problems are usually solved recursively. *

- ( ) False
- (•) True

Will be reviewed

10

The worst-case time complexity of Quick Sort is: *

- ( ) $O(n)$
- ( ) $O(\lg n)$
- ( ) $O(n \lg n)$
- (•) $O(n^2)$

Will be reviewed

11

Priority Queue usually implemented with heap structure *

- (•) True

- [ ] False

Will be reviewed

12

Let T(n) be defined by T(1) = 7, and T(n+1) = 3n + T(n), for all integers n ≥ 1. Which of the following represents the order of growth of T(n) as a function of n? *

- [x] $O\left(n^2\right)$
- [ ] $O\left(n \lg n\right)$
- [ ] $O\left(n\right)$
- [ ] $O\left(n^2 \lg n\right)$

Will be reviewed

13

Consider the shown graph. The outdegree for the vertex D is: *

- [ ] 2
- [ ] 3
- [ ] 1
- [x] 0

Will be reviewed

14

A minimum spanning tree of a given graph may contain a cycle edge. *

- ○ True
- ◉ False

Will be reviewed

15

Consider the shown graph. The indegree for the vertex D is: *

- ◉ 0
- ○ 2
- ○ 1
- ○ 3

Will be reviewed

16

For undirected graphs, the adjacency matrix is not symmetric *

- ◉ False
- ○ True

Will be reviewed

17

Form these edge set, E = {(1,2), (1,4), (1,5), (2,3), (2,5), (3,2), (4,5)}, This can set represent a directed graph or undirected graph
*

- (●) False
- ( ) True

Will be reviewed

18

The essential part of Heap sort is construction of max-heap. Consider the tree shown below, the node 24 violates the max-heap property. Once heapify procedure is applied to it, which position will it be in?
*

- ( ) 8
- ( ) 5
- (●) 9
- ( ) 4

Will be reviewed

19

The worst-case time complexity of Merge Sort is:
*

- (●) $O\ (n\ \lg n)$
- ( ) $O\ (n)$

- $O(\lg n)$
- $O(n^2)$

Will be reviewed

20

Which of the following is **false** in the case of a spanning tree of a graph G?
*

- It is tree that spans G
- It is a subgraph of the G
- It can be either cyclic or acyclic
- It includes every vertex of the G

Will be reviewed

21

Consider the following heap after build heap phase. What will be its corresponding array?
*

- 97,53,59,26,41,58,31
- 26,31,41,53,58,59,97
- 26,41,53,97,31,58,59
- 26,53,41,97,58,59,31

Will be reviewed

22

If we have the next undirected graph { (a,b,3), (b,c,8), (c,d,2), (d,e,5), (e,a,14), (e,b,10) }, then we apply Kruskal algorithm on that graph. the first 3 edges that will be picked (in order) will be …… *

- (b, c), (d, e), (e, b)
- (c, d), (a, b), (d, e)
- (a, b), (c, d), (e, a)
- (e, b), (b, c), (c, d)

Will be reviewed

23

Consider the shown graph. Using Kruskal's algorithm, which edge will be selected first? *

- AC
- CB
- CF
- CD

Will be reviewed

24

Consider the shown graph. Which of the following edges cannot be found in the minimum spanning tree of this graph? *

- [ ] CD
- [ ] CF
- [ ] CB
- [x] AC

Will be reviewed

25

Given a binary-max heap. The elements are stored in an arrays as 25, 14, 16, 13, 10, 8, 12. What is the content of the array after two delete operations? *

- [ ] 14,13,12,10,8
- [ ] 14,12,13,10,8
- [ ] 14,13,8,12,10
- [x] 14,13,12,8,10

Will be reviewed

26

Consider the shown graph. Using Dijkstra's Algorithm, the shortest path starting from A to E will costs: *

- [ ] 8
- [x] 6
- [ ] 5

- ○ 7

Will be reviewed

27

The adjacency matrix of an undirected graph is asymmetric *

- ◉ False
- ○ True

Will be reviewed

28

If we have the next undirected graph { (a,b,3), (b,c,8), (c,d,2), (d,e,5), (e,a,14), (e,b,10) }, then we apply Prime algorithm on that graph. After 3 queue extraction within the algorithm, the vertices in the queue will be …… *

- ○ e, b
- ○ d, c
- ○ c, b
- ◉ e, d

Will be reviewed

29

By applying the quick sort on the next array, 56 | 82 | 92 | 74 | 76 | 68 | 88 | 68 | 48 | 74 | 18 | 94 , after the second round of the partitioning the array will be like: *

- ○ 56 | 82 | 92 | 74 | 76 | 68 | 88 | 68 | 48 | 74 | 18 | 94

- ○ 48 | 18 | 56 | 74 | 76 | 68 | 88 | 68 | 92 | 74 | 82 | 94
- ○ 56 | 18 | 48 | 74 | 76 | 68 | 88 | 68 | 92 | 74 | 82 | 94
- ◉ 18 | 48 | 56 | 74 | 76 | 68 | 88 | 68 | 92 | 74 | 82 | 94

Will be reviewed

30

What is the weight of the minimum spanning tree for the given graph using Kruskal's algorithm?
*

- ◉ 14
- ○ 15
- ○ 16
- ○ 12

Will be reviewed

31

Greedy algorithms look for future benefit do not choose the next piece of the solution that offers the most obvious and immediate benefit
*

- ○ True
- ◉ False

Will be reviewed

32

For any directed graph *

$\sum u \epsilon v \deg(v) = 2 \mid E|$

- True
- False

Will be reviewed

33

Form these edge set, E = {(1,2), (1,4), (1,5), (2,3), (2,5), (3,2), (4,5)}, This can set represent a directed graph.
*

- True
- False

Will be reviewed

34

Which of the following array representation is CORRECT for the max heap of the following data elements? 20, 30, 70, 10, 5, 50, 80 *

- 5, 10, 20, 30, 50, 70, 80
- 70, 20, 80, 10, 5, 30, 50
- 80, 20, 70, 10, 5, 30, 50
- 80, 70, 50, 30, 20, 10, 5

Will be reviewed

35

The next adjacency matrix represents *

- (●) undirected graph
- ( ) directed graph

Will be reviewed

36

The Θ expression (tight bound) to describe the number of operations required for the following algorithms is ……..:

```
for i = 1 to A.length - 1
  for j = A.length downto i + 1
     if A[j] < A[j-1]
        exchange A[j] with A[j-1] *
```

- ( ) $\Theta(n^2 lgn)$
- ( ) $\Theta(2^n)$
- (●) $\Theta(n^2)$
- ( ) $\Theta(n)$

Will be reviewed

37

By applying the construct heap algorithm on the next array, 15 | 19 | 10 | 7 | 17 | 16 ,the resulting array will be *

- [ ] 19 | 17 | 16 | 15 | 7| 10
- [ ] 15 | 19 | 10 | 7 | 17 | 16
- [x] 19 | 17 | 16 | 7 | 15| 10
- [ ] 19 | 15 | 10 | 7 | 17 | 16

Will be reviewed

38

Consider the shown graph. Using Dijkstra's Algorithm, the shortest path starting from A to D will costs: *

- [ ] 6
- [ ] 7
- [x] 5
- [ ] 8

Will be reviewed

39

Merge sort make sorting in place *

- [x] False
- [ ] True

Will be reviewed

40

*

$T(n) = T(n/6) + 2T(n/6) + 3T(n/6) + n^2, \qquad T(n) = O(\ldots\ldots)$

- $\lg\ n$
- $n \lg n$
- $Option2$
- n^2

Will be reviewed

41

For any undirected graph *

$\sum u \epsilon v \deg(v) = 2 \mid E|$

- $False$
- $True$

Will be reviewed

42

The Kruskal's algorithm can also run on the disconnected graphs.
*

- True
- False

Will be reviewed

43

By applying the counting sort on the next array, 4 | 1 | 3 | 4 | 3, the count array C will be like (before cumulative summation): *

- (●) 0 | 1 | 0 | 2 | 2
- ( ) 0 | 1 | 1 | 2 | 4
- ( ) 0 | 0 | 1 | 1 | 3
- ( ) 0 | 1 | 1 | 3 | 5

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